KESTREL LABS

Unit 4, Harrow Works
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Q3 Test Programme Report

Autonomous pallet mover, firmware 4.2

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Summary

The programme ran 36 test scenarios between July and September. Overall pass rate rose to 93.4%, up from 88.1% in Q2. Two scenarios still fail more than one run in five and are carried into Q4.

> **Note**
> Pass rates count a run as passed only if the vehicle completes the route with no operator intervention.

Method

1. Each scenario runs at least 20 times on firmware 4.2.
2. Runs are split across the three indoor bays and the outdoor yard.
3. Failures are logged with sensor traces and reviewed within 48 hours.

Results by scenario

Scenario	Location	Runs	Passed	Pass rate
KT-001	Bay 2	27	24	88.9%
KT-002	Bay 3	34	32	94.1%
KT-003	Outdoor yard	41	40	97.6%
KT-004	Bay 1	23	23	100.0%
KT-005	Bay 2	30	27	90.0%
KT-006	Bay 3	37	35	94.6%
KT-007	Outdoor yard	44	43	97.7%
KT-008	Bay 1	26	26	100.0%
KT-009	Bay 2	33	30	90.9%
KT-010	Bay 3	40	38	95.0%
KT-011	Outdoor yard	22	21	95.5%
KT-012	Bay 1	29	29	100.0%
KT-013	Bay 2	36	33	91.7%

Scenario	Location	Runs	Passed	Pass rate
KT-014	Bay 3	43	41	95.3%
KT-015	Outdoor yard	25	24	96.0%
KT-016	Bay 1	32	32	100.0%
KT-017	Bay 2	39	36	92.3%
KT-018	Bay 3	21	19	90.5%
KT-019	Outdoor yard	28	27	96.4%
KT-020	Bay 1	35	35	100.0%
KT-021	Bay 2	42	39	92.9%
KT-022	Bay 3	24	22	91.7%
KT-023	Outdoor yard	31	30	96.8%
KT-024	Bay 1	38	38	100.0%
KT-025	Bay 2	20	17	85.0%
KT-026	Bay 3	27	25	92.6%
KT-027	Outdoor yard	34	33	97.1%
KT-028	Bay 1	41	41	100.0%
KT-029	Bay 2	23	20	87.0%
KT-030	Bay 3	30	28	93.3%
KT-031	Outdoor yard	37	36	97.3%
KT-032	Bay 1	44	44	100.0%
KT-033	Bay 2	26	23	88.5%
KT-034	Bay 3	33	31	93.9%
KT-035	Outdoor yard	40	39	97.5%
KT-036	Bay 1	22	22	100.0%

Carried into Q4

- KT-012: wet-floor braking distance in Bay 1
- KT-027: low-sun glare on the outdoor yard camera

> **Warning**
> Do not deploy firmware 4.2 outdoors in low sun until KT-027 is resolved.